

Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment

MLOps_Assignment_2

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Training and Fine-Tuning BERT for Goodreads Review Classification

Classifying Goodreads Reviews By Book Genre By Maria Antoniak, Melanie Walsh, and the AI for Humanists Team Updated: 2024-11-05

This project demonstrates an end-to-end machine learning workflow including model training, experiment tracking, model versioning, and inference using Hugging Face Transformers, Weights & Biases, and Kaggle.

A BERT model is fine-tuned on Goodreads reviews from the UCSD Book Graph with the goal of predicting the genre of the book being reviewed.

The genres include:

- poetry
- comics & graphic
- fantasy & paranormal
- history & biography
- mystery, thriller, & crime
- romance
- young adult

- This project focused on fine-tuning a BERT-based transformer model for text classification using Goodreads book reviews.
- The objective was to classify reviews according to book genres by leveraging transfer learning from a pretrained language model.
- The workflow included configuring a Kaggle Notebook environment, enabling GPU acceleration, setting up experiment tracking with Weights & Biases (W&B;), and publishing the final trained model to Hugging Face.
- The project emphasized reproducibility, experiment management, and practical MLOps workflow integration in addition to model performance.

Basic steps involved in using BERT and HuggingFace:

1. Divide your data into training and test sets.
 2. Encode your data into a format BERT will understand.
 3. Combine your data and labels into dataset objects.
 4. Load the pre-trained BERT model.
 5. Fine-tune the model using your training data.
 6. Predict new labels and evaluate performance on your test data.
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Why BERT Model -

BERT (Bidirectional Encoder Representations from Transformers) was selected because it is one of the most widely used transformer architectures for natural language processing tasks. Unlike traditional machine learning approaches such as TF-IDF with Logistic Regression, BERT captures contextual meaning from both left and right directions in a sentence. This makes it highly effective for understanding the semantics of Goodreads reviews, where meaning often depends on context, tone, and sentence structure. The Hugging Face Transformers library was used because it provides pretrained checkpoints, tokenizers, and training utilities that simplify the fine-tuning workflow. Fine-tuning a pretrained BERT model significantly reduces training time compared to training a model from scratch while still achieving strong performance on classification tasks. The model was also chosen because it integrates smoothly with modern MLOps tools including: Kaggle for cloud-based GPU training Weights & Biases for experiment tracking Hugging Face Hub for model hosting and inference Another reason for selecting BERT was its strong community support and extensive documentation. This reduced setup complexity and made debugging easier during implementation

Kaggle training setup — how did you configure Kaggle Notebook, GPU, and Secrets?

The project was implemented using a Kaggle Notebook environment because Kaggle provides free access to GPUs and supports seamless notebook-based experimentation.

The notebook was configured with GPU acceleration enabled from the Kaggle settings panel. A T4 GPU was used during training to speed up tokenization and fine-tuning operations.

The training environment included: Python with Hugging Face Transformers PyTorch backend Datasets library for loading and preprocessing data Weights & Biases (W&B;) for logging metrics To securely

connect W&B; and Hugging Face accounts, Kaggle Secrets were configured. API tokens were stored as notebook secrets instead of hardcoding them into scripts.

This improved security and followed good MLOps practices.

The workflow included the following steps:

Load Goodreads review dataset
Clean and preprocess text data
Tokenize reviews using the BERT tokenizer
Fine-tune the pretrained model
Track training metrics with W&B;

Save and upload the final model to Hugging Face
Create an inference script for prediction testing .

The notebook also generated a requirements.txt file and GitHub repository structure to support reproducibility and submission requirements.

Challenges & learnings — what was hard? What would you do differently?

One of the biggest challenges during the project was configuring the training environment correctly across Kaggle, Hugging Face, and W&B.;

Issues related to authentication tokens, public/private visibility settings, and package installation errors required troubleshooting.

Network-related pip installation problems and GPU session resets on Kaggle occasionally interrupted training.

Another challenge involved understanding transformer workflows compared to traditional machine learning pipelines.

Tokenization, batching, padding, and GPU memory management introduced additional complexity. Fine-tuning transformer models also required careful parameter selection such as learning rate, batch size, and number of epochs.

The project provided several important learnings: How transfer learning improves NLP performance
How to use GPUs efficiently for deep learning
How experiment tracking improves reproducibility
How to structure an end-to-end MLOps workflow
How to publish and share machine learning artifacts publicly
If the project were repeated, several improvements could be made.

More extensive hyperparameter tuning could improve performance further.

Additional preprocessing techniques such as removing noisy reviews or balancing classes could also help.

Using a larger validation strategy or cross-validation would provide more reliable evaluation metrics. Overall, the project was valuable for understanding the complete lifecycle of modern NLP model development, from training and evaluation to deployment and reproducibility.

Conclusion

This project successfully demonstrated the fine-tuning of a BERT-based model for Goodreads review classification using Kaggle infrastructure and modern MLOps tools. The workflow combined model training, experiment tracking, secure configuration management, and reproducible deployment practices. Beyond achieving classification results, the project highlighted the importance of workflow organization, automation, and collaboration tools in machine learning engineering.

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